



\*\*\*spare part\*\*\* SIMATIC S7-300, digital input SM 321, isolated 32 DI, 24 V DC, 1x 40-pole

| General information                                             |             |
|-----------------------------------------------------------------|-------------|
| Product function                                                |             |
| • Isochronous mode                                              | No          |
| Supply voltage                                                  |             |
| Load voltage L+                                                 |             |
| • Rated value (DC)                                              | 24 V        |
| • permissible range, lower limit (DC)                           | 20.4 V      |
| • permissible range, upper limit (DC)                           | 28.8 V      |
| Input current                                                   |             |
| from backplane bus 5 V DC, max.                                 | 15 mA       |
| Power loss                                                      |             |
| Power loss, typ.                                                | 6.5 W       |
| Digital inputs                                                  |             |
| Number of digital inputs                                        | 32          |
| Input characteristic curve in accordance with IEC 61131, type 1 | Yes         |
| Number of simultaneously controllable inputs                    |             |
| horizontal installation                                         |             |
| — up to 40 °C, max.                                             | 32          |
| — up to 60 °C, max.                                             | 16          |
| vertical installation                                           |             |
| — up to 40 °C, max.                                             | 32          |
| Input voltage                                                   |             |
| • Type of input voltage                                         | DC          |
| • Rated value (DC)                                              | 24 V        |
| • for signal "0"                                                | -30 to +5 V |
| • for signal "1"                                                | 13 to 30V   |
| Input current                                                   |             |
| • for signal "1", typ.                                          | 7 mA        |
| Input delay (for rated value of input voltage)                  |             |
| for standard inputs                                             |             |
| — parameterizable                                               | No          |
| — at "0" to "1", min.                                           | 1.2 ms      |
| — at "0" to "1", max.                                           | 4.8 ms      |
| — at "1" to "0", min.                                           | 1.2 ms      |
| — at "1" to "0", max.                                           | 4.8 ms      |
| Cable length                                                    |             |
| • shielded, max.                                                | 1 000 m     |
| • unshielded, max.                                              | 600 m       |
| Encoder                                                         |             |

|                                                       |                  |
|-------------------------------------------------------|------------------|
| Connectable encoders                                  |                  |
| • 2-wire sensor                                       | Yes              |
| — permissible quiescent current (2-wire sensor), max. | 1.5 mA           |
| Interrupts/diagnostics/status information             |                  |
| Alarms                                                | No               |
| Diagnostics function                                  | No               |
| Alarms                                                |                  |
| • Diagnostic alarm                                    | No               |
| • Hardware interrupt                                  | No               |
| Diagnostics indication LED                            |                  |
| • Status indicator digital input (green)              | Yes              |
| Potential separation                                  |                  |
| Potential separation digital inputs                   |                  |
| • between the channels                                | No               |
| • between the channels, in groups of                  | 16               |
| • between the channels and backplane bus              | Yes; Optocoupler |
| Isolation                                             |                  |
| Isolation tested with                                 | 500 V DC         |
| Connection method                                     |                  |
| required front connector                              | 40-pin           |
| Dimensions                                            |                  |
| Width                                                 | 40 mm            |
| Height                                                | 125 mm           |
| Depth                                                 | 120 mm           |
| Weights                                               |                  |
| Weight, approx.                                       | 260 g            |
| Classifications                                       |                  |

|        | Version | Classification |
|--------|---------|----------------|
| eClass | 14      | 27-24-22-04    |
| eClass | 12      | 27-24-22-04    |
| eClass | 9.1     | 27-24-22-04    |
| eClass | 9       | 27-24-22-04    |
| eClass | 8       | 27-24-22-04    |
| eClass | 7.1     | 27-24-22-04    |
| eClass | 6       | 27-24-22-04    |
| ETIM   | 10      | EC001419       |
| ETIM   | 9       | EC001419       |
| ETIM   | 8       | EC001419       |
| ETIM   | 7       | EC001419       |
| IDEA   | 4       | 3566           |
| UNSPSC | 15      | 32-15-17-05    |

|                          |  |
|--------------------------|--|
| Approvals / Certificates |  |
| General Product Approval |  |



[Miscellaneous](#)



|                          |     |                                |
|--------------------------|-----|--------------------------------|
| General Product Approval | EMV | For use in hazardous locations |
|--------------------------|-----|--------------------------------|

[China RoHS](#)



[FM](#)

|                                |                      |
|--------------------------------|----------------------|
| For use in hazardous locations | Maritime application |
|--------------------------------|----------------------|



[Miscellaneous](#)

[CCC-Ex](#)



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| Maritime application |
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[NK / Nippon Kaiji Kyokai](#)



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| Maritime application |
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[CCS \(China Classification Society\)](#)

last modified:

4/7/2025 